

Sales Order Data Analysis

A Multi-Tool Data Analysis Project
Using Excel, Python, SQL & Power BI

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May, 2026

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Project Overview

Objective

To Clean and analyze a sales order dataset using 4 professional tools

Tools Used

- Microsoft Excel
- Python (Jupyter)
- SQL
- Power BI

Dataset

- 1,200 rows
- 14 Columns
- Sales order data
- Year 2023

Outcome

- Cleaned dataset
- Full analysis
- Dashboard built
- Portfolio created

About the Dataset

1,200

Rows of Data

14

Columns

2023

Period

Data Columns:

·OrderID ·CustomerID ·Date ·Product ·Quantity ·UnitPrice ·TotalPrice ·PaymentMethod ·OrderStatus ·CouponCode
·TrackingNumber ·ShippingAddress ·ReferralSource

Data Issues Found:

·309 missing CouponCode values ·Floating point noise in price columns ·Date formatting issues

Data Cleaning – Microsoft Excel

Step 1 – Created working copy to preserve original data

Step 2 – Formatted Date column from serial numbers to DD/MM/YYYY

Step 3 – Rounded UnitPrice & TotalPrice to 2 decimal places

Step 4 – Filled 309 blank CouponCode cells with “No Coupon”

Step 5 – Verified zero duplicates & validated TotalPrice accuracy

Result: 1,200 rows fully cleaned – zero errors found!

Data Cleaning – Python (Jupyter Notebook)

Libraries Used

- Pandas
Data manipulation & Cleaning
- Matplotlib
Data visualization & charts
- Seaborn
Statistical visualization
- OpenPyXL
Excel file handling

Cleaning Steps

- Loaded dataset with Pandas
- Fixed Date column format
- Rounded price decimal
- Filled 309 blank CouponCodes
- Stripped hidden spaces
- Validated TotalPrice accuracy
- Exported as Excel & CSV

```
df['CouponCode'] = df['CouponCode'].fillna("No Coupon")  
df['UnitPrice'] = df['UnitPrice'].round(2)  
df.to_excel("Cleaned_Dataset.xlsx", index=False)
```

Result: Cleaned dataset exported as Excel & CSV successfully!

Data Cleaning – SQL & Power Query

SQL (SSMS)

- Checked 1,200 total rows
- Found 309 missing CouponCodes
- Filled blanks with “No Coupon”
- Verified zero duplicates
- Trimmed all text columns
- Rounded price columns
- Validated TotalPrice accuracy

Power Query (Power BI)

- Renamed query to Orders_Cleaned
- Promoted headers correctly
- Fixed all data types
- Removed duplicate rows
- Filled blank CouponCodes
- Trimmed all text columns
- Rounded price columns

```
UPDATED [dbo].[Sheet1$] SET CouponCode = 'No Coupon' WHERE CouponCode IS NULL
--(309 row affected)
```

Result: Data Cleaning in both SQL & Power Query – Zero errors found!

Data Analysis – Key Results

\$1.26M

Total Revenue

1,200

Total Orders

Chair

Top Product

June

Best Month

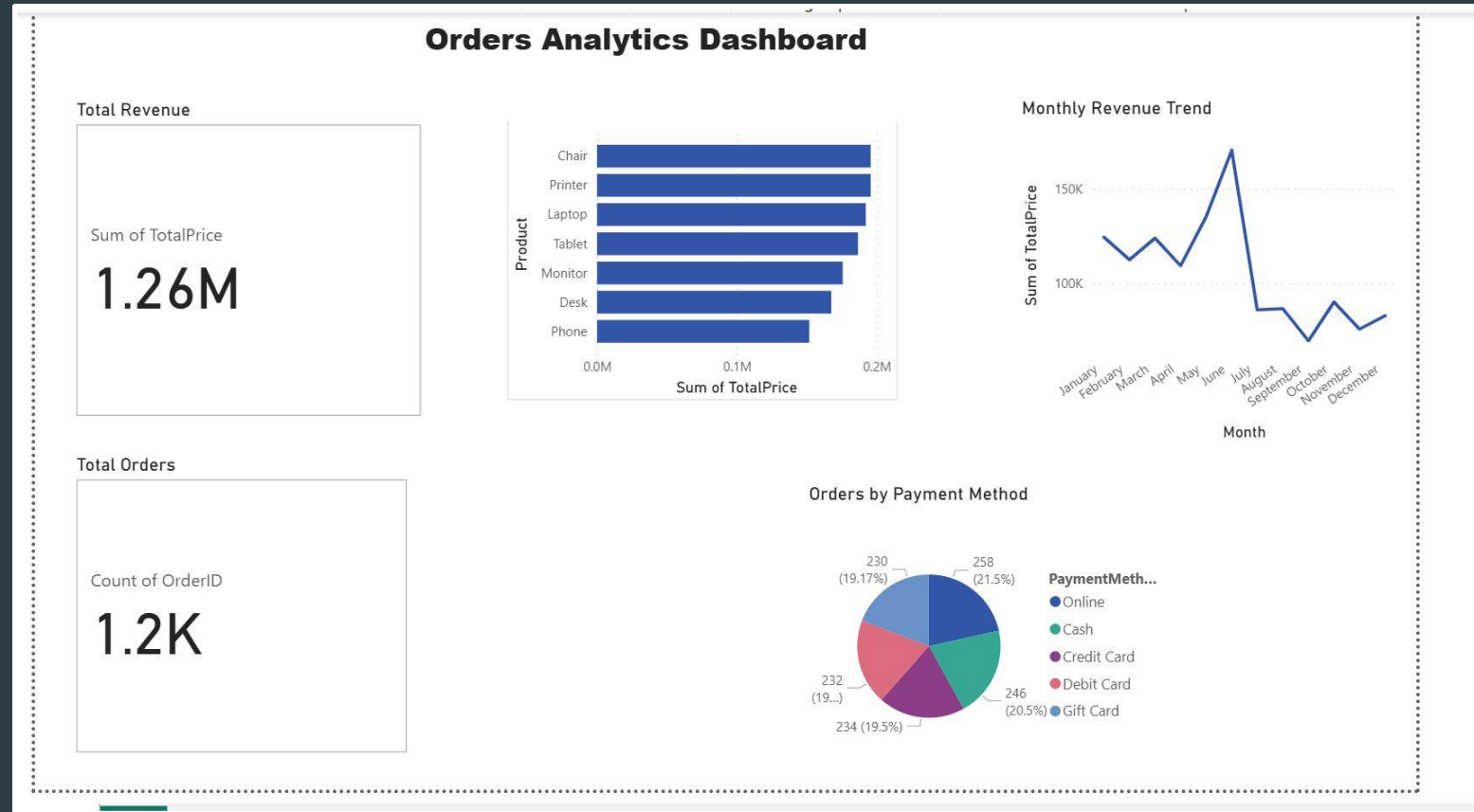
Excel PivotTables

Python Pandas & Matplotlib

SQL Queries

Power BI Dashboard

Power BI Dashboard



Dashboard Contains:

- ✓ Total Revenue Card
- ✓ Total Orders Card
- ✓ Sales by product
- ✓ Orders by Payment
- ✓ Monthly Revenue Trend

Total Revenue: \$1.26M across 1,200 orders – Built with Power BI Desktop

Descriptive Statistics

Statistical summary of TotalPrice & Quantity columns

Total Price Statistics

\$1,050.00

Mean

\$1,048.50

Median

\$10.20

Minimum

\$3,456.40

Maximum

Quantity Statistics

3.00

Mean

3.00

Median

1

Minimum

5

Maximum

Key Insight: Mean $\approx$ Median for TotalPrice (\$1,050 vs \$1,048) indicating a fairly symmetrical distribution with no significant skew!

Data is normally distributed – Mean & Median are almost identical!

Outlier Analysis

Identifying unusual orders using the IQR Method

\$-1,341

Lower Boundary

\$3,330.41

Upper Boundary

8

Outliers Found

OrderID	Product	Quantity	UnitPrice	TotalPrice
ORD200328	Tablet	5	\$674.04	\$3,370.20
ORD200107	Printer	5	\$670.75	\$3,353.75
ORD200632	Laptop	5	\$678.16	\$3,390.80
ORD201122	Monitor	5	\$678.19	\$3,390.95
ORD200789	Tablet	5	\$691.28	\$3,456.40
ORD201065	Printer	5	\$666.80	\$3,334.00
ORD200469	Chair	5	\$676.98	\$3,384.90
ORD200326	Laptop	5	\$670.48	\$3,352.40

Key Observations:

- All 8 outliers have Quantity = 5 (bulk orders)
- All are HIGH value orders
-Not data errors

Products involved:

Tablet x 2
Printer x 2
Laptop x 2
Monitor x 1
Chair x 1

8 High-Value Bulk Orders Identified – All Legitimate, No Data Errors Found!

Key Findings & Recommendations

Key Findings

Total Revenue: \$1.26M
generated across 1,200 orders

Chair is the best selling product
by revenue.

June recorded the highest
revenue & quantity sold.

All 5 payment methods are
evenly distributed (20% each).

Clear top 10 high value
customer identified.

Recommendations

Stock up on Chairs & Printers –
top revenue drivers.

Investigate the June spike and
replicate the strategy.

Promote Phone sales through
product bundles & discounts.

Reward top 10 customers with
loyalty programs.

Support all payment methods to
avoid losing sales.

Prepared by Ifeoma Precious Ugwu

Tools: ·Excel ·Python ·SQL ·Power BI